Edge Effect Issue on Hyperspectral LiDAR Point Cloud Data: Generation Principle and Framework of Detection and Filtering

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***Abstract*—The edge effect, caused by partial illumination of leaf boundaries, leads to significant hyperspectral LiDAR (HSL) echo-intensity loss and severely restricts the precise expression of the spectral characteristics and retrieval accuracy of biochemical components in the leaf-edge area. Despite being one of the major radiometric effects in LiDAR sensing, the edge effect has not been systematically investigated for hyperspectral intensity data. Along these lines, this study presents the first comprehensive exploration, detection, and filtering of edge effects in HSL-collected data. We analyze the physical mechanisms responsible for multilayer edge-effect formation and propose a new integrated detection and filtering framework, incorporating a histogram of intensity distribution, edge detection, and spherical-spatial filtering (HIDEDaSF). Results obtained from the HSL system on broadleaf plants demonstrate that our framework robustly detects edge-affected spectral points across 32 wavelengths. After filtering and correction, the standard deviation and coefficient of variation (CV) of edge-region intensities are reduced by 22.68% and 28.30%, respectively, and the mean ratio of CV is 0.7288, less than 1, confirming the stability and effectiveness of the proposed algorithm framework again. The HIDEDaSF framework preserves valid spectral information and significantly improves the consistency and quantitative reliability of hyperspectral intensity data at leaf-edge zones. Although tailored for the HSL system, this framework is also adaptable to other multi and HSL platforms. This study enriches the fundamental knowledge of edge-effect radiometric behaviors and expands the methodology foundation for fine-scale HSL vegetation remote sensing. We are willing to make our codes freely available via https://github.com/Jie-Bai/HIDEDaSF**

***Index Terms*—Edge effect, edge detection, HIDEDaSF, hyperspectral LiDAR, hyperspectral point cloud, intensity density, intensity filtering, spherical-spatial filtering.**

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I. Introduction

EVER since the emergence of light detection and ranging (LiDAR) sensors with 3-D imaging capability, the way of viewing targets on Earth has revolutionized, and the remote sensing community has valued fully 3-D structural data obtained from LiDAR sensors over the past two decades [1], [2]. Compared to optical 2-D imagery, LiDAR supplies the researchers with greater potential for accurate vertical description and mapping for the spatial property of targets via 3-D coordinates [3]. Besides the 3-D coordinates, most LiDAR sensors provide another attribute called intensity for each point in the corresponding point cloud [4]. This attribute is usually closely related to the physical and chemical properties of the target and is used for visualization. Nowadays, the intensity attribute of LiDAR has been widely applied in various geoscience scenes, as an essential attribute about surface physical properties [5]. In particular, intensity information can be applied to land cover classification [6], vegetation species identification [7], mineral classification [8], [9], atmospheric parameter detection [10], [11], water content inversion [4], chlorophyll parameter inversion [12], [13], among others. Recent research studies demonstrate that intensity information at different wavelengths can be used for quantitative remote sensing studies of the physical properties of geoscientific targets [5], [14], [15]. As intensity information is increasingly utilized, a proper understanding and quantitative processing of intensity data is one of the most crucial steps in further treatment and interpretation for a 3-D point cloud.

Generally, the raw intensity data recorded by LiDAR systems are strongly influenced by several radiometric factors [16], [17], [18], [19], [20], [21], primarily comprising the range effect, the directional reflection effect, and the edge effect. Bai et al. [14] theoretically demonstrated the

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independence between the range effect and the directional reflection effect, and further established the first hyperspectral LiDAR (HSL) physical equation that accounts for the directional heterogeneity of the target. In research [14], the wavelength dependence of these effects and the mainstream correction algorithms were also systematically summarized [14]. By contrast, few comprehensive investigations, analyses, and algorithmic developments have been conducted to address the edge effect on intensity data [22]. The edge effect, also known as the subspot/subfootprint effect, refers to the phenomenon that when multiple objects with various natural reflectance properties are simultaneously located at a beam path and illuminated within a single laser footprint along the beam path, the echo energy of the specially focused object within the laser spot will be lost at some extent. In such cases, the backscattered energy from either the foreground or background target will be significantly reduced, which will be clearly visible on the recorded waveform. This phenomenon typically arises along the structural boundaries of objects in 3-D space. Since LiDAR intensity is closely associated with the physiological and biochemical properties of the scanned target [15], the edge effect may introduce substantial uncertainties into the retrieval of these properties at edge zones. Thereby, the presence of edge effects severely hinders the accurate representation of the 3-D physiological and biochemical characteristics of terrestrial targets.

HSL represents a new generation of active optical remote sensing technology in the aerospace domain, capable of simultaneously acquiring the geometric structure and vertical spectral information of vegetation [8], [12], [13], [23], [24]. This unique capability provides a novel approach for retrieving the spatial distribution of biochemical parameters and conducting 3-D health monitoring [24]. The small-footprint hyperspectral laser can precisely infer the biochemical parameters within the laser footprint through the intensity and deuterogenic reflectance value. Nevertheless, when the laser beam illuminates the edge of a leaf, both the leaf and its surrounding background share the incident energy, leading to a substantial loss in the leaf's reflected return signal. When multiple objects with different reflectance properties coexist within a single footprint, the hyperspectral point clouds exhibit pronounced edge effects, manifested as lower intensity values along leaf margins compared to their interior. A severe nonlinear distortion in the spectral signatures of the edge points would occur. Such complex edge effects distort the 3-D spectral information and markedly reduce its ability to precisely characterize leaf biochemical contents, and thus severely constrain its applicability in precise vegetation remote sensing. Under complex 3-D canopy environments, investigating the detection and filtering of vegetation edge effects in active hyperspectral point clouds is of great theoretical and practical significance for 3-D vegetation health sensing.

For the complexity of this physical process for edge effect, few studies have been tailored for edge-effect modeling and correction, and instead, they commonly adopt provisional strategies by ignoring or excluding. With the introduction of hyperspectral point clouds, applications targeting vegetation biochemical parameters have clearly revealed pronounced biochemical anomalies within the edge-affected regions [15], [25]. Previously, in HSL applications on vegetation, edge-affected point cloud intensities were directly used for inversion modeling without filtering or correction, or edge points were simply discarded as invalid observations during data analysis [25], [26]. Such treatments, however, would cause a twofold dilemma: first, when distorted spectral intensities are directly involved in parameter inversion, systematic biases are inevitably introduced, leading to large uncertainties in the estimation of key biochemical traits such as chlorophyll and nitrogen. Thereby, it weakens the quantitative retrieval capability of HSL for vegetation parameters. For the second way, although removing edge points prevented error propagation, it inevitably results in the permanent loss of valuable data in edge regions where early biological or nonbiological stress may happen, and that contain the most sensitive zone for physiological change. Some other studies have attempted to mitigate edge effects indirectly by combining the multiple-wavelength backscattered intensity or reflectance into spectral indices [22]. Nevertheless, such spectral indices essentially represent derivative variables of raw return intensity and may not be meaningful optical indicators sensitive to key vegetation biochemical constituents such as chlorophyll or nitrogen content. Therefore, spectral index-based approaches severely limit the potential of 3-D mapping for vegetation physiological parameters. Consequently, the aforementioned three ways of ignoring, data omission, as well as spectral index, cannot fundamentally resolve the edge-effect problem, and it underscores the urgent need to develop dedicated detection and filtering algorithms for edge effect on vegetation hyperspectral point cloud data.

Along these lines, this study carried out the first comprehensive exploration, analysis, and filtering of edge effects on hyperspectral intensity data. The overall objectives of this article are follows: 1) to analyze the intensity characteristics and differences of vegetation point clouds at different wavelengths; 2) to develop an universal edge-effect detection and filtering algorithm for HSL sensor; 3) to quantitatively analyze the changes for intensity distribution of the vegetation hyperspectral point cloud with edge-effect filtering; and 4) to examine accuracy of the edge-effect filtering algorithm using standard deviation (σ), coefficient of variation (CV), and ratio of CV (ε).

II. Data and Methodology

A. Experiment Installation

The 3-D hyperspectral intensity data of the vegetation were acquired using an HSL system developed by us (Fig. 1) [1]. This HSL system is capable of simultaneously capturing backscattered signals over the spectral range of 409–914 nm, divided into 32 discrete wavelengths. The spectral resolution of the system is 16 or 17 nm, and the laser beam divergence is 5 mrad. The specific wavelengths can be found in a previous study [1]. The emitted laser pulse has a collimated direction with a Gaussian waveform shape, while the backscattered signal is dispersed by a diffraction grating to obtain 32-wavelength hyperspectral waveform measurements. Each

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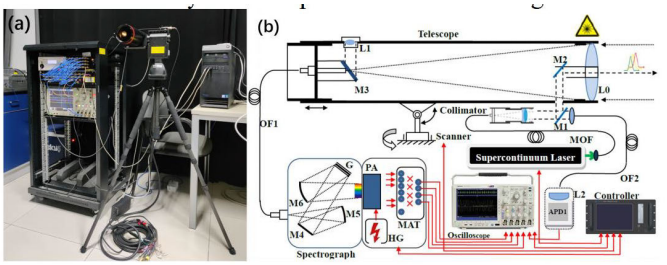


Fig. 1. HSL prototype with a supercontinuum pulsed laser source. (a) Shot of the prototype. (b) Optical structure. L0, L1, and L2 represent the achromatic lens. M1 represents the microlens. M2 and M4 represent the reflectors. M3 represents the diffuse mirror. M5 represents the collimation mirror. M6 represents the converging mirror. G represents the grating spectrometer. APD1 represents an avalanche photodiode. MAT represents the switch matrix. OF1 and OF2 represent the optical fiber. MOF represents the microstructure optical fiber. PA represents a photosensitive array. HG represents the generator.



Fig. 2. 3-D scanning photograph of *Alocasia macrorrhiza*.

wavelength's waveform records the cumulative backscattered energy along the laser path within a single footprint, corresponding to returns of the target surface from different heights. The hyperspectral laser scanning setup was performed using a FLIR PTU-D48E pan-tilt platform to achieve full 3-D scanning of the target, resulting in the generation of hyperspectral point clouds. A detailed schematic of the system components is shown in Fig. 1.

B. Data Acquisition

Through the rotation of the pan-tilt unit, a 3-D hyperspectral point cloud of *Alocasia macrorrhiza* plant was acquired. *Alocasia macrorrhiza* is a broad-leaved plant, with the individual in this study having a height of approximately 90 cm and a width of 80 cm (Fig. 2). The experiment was conducted indoors under ambient light conditions. Since the laser pulse has a relatively high energy with a peak power of 20 kW, the ambient illumination seldom owns influence on the scanning results, and it is negligible. The laser scanner was positioned about 5 m from the target, and a total of 9552 HSL points was recorded on the plant. Fig. 3 presents the HSL-collected backscattered intensity across various spectral wavelengths. As shown in Fig. 3, the intensity characteristics vary significantly among different wavelengths, demonstrating the HSL system's capability to capture wavelength-dependent reflectance features of the target.

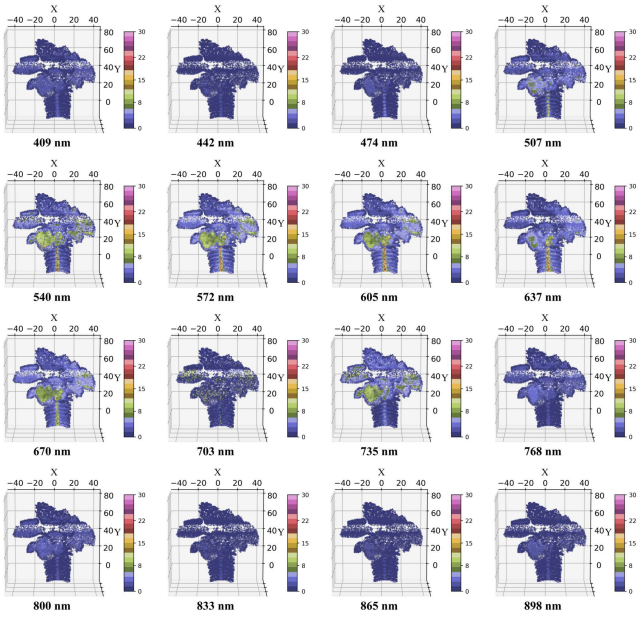


Fig. 3. HSL-collected backscattered intensity across different spectral wavelengths (unit: mV). The reason for choosing the colormap "tab20b" is to differentiate and visualize in detail the intensity and the edge effect for leaves.

As shown in Fig. 3, distinct differences in backscattered intensity are observed between the leaf-edge regions and the central portions of the leaves. The echo intensity at the leaf edges is significantly lower than that in the interior regions. Detecting and filtering these edge-affected areas constitutes the primary task in this study. Although for some wavelengths, such as 442 and 703 nm, the leaf edges are not visually apparent due to low signal-to-noise ratio (SNR) [1], the edge effect is clearly observable at other wavelengths. This phenomenon further highlights the spectral advantages of the HSL system over single-wavelength LiDAR and provides new insights for the design of our edge-detection algorithm. In the lower part of the plant, the pot exhibits high brightness in the central area due to its ceramic material and direct orientation toward the laser beam, which results in strong backscattering. In fact, it can be observed that only the leaf on the far left is somewhat hollow. This is due to the size of the laser spot of the instrument, the density of the points, as well as the weak reflection signal of the leaf. In the other parts of the plant leaves, the white areas are actually the physical gaps between the leaves, not holes.

C. Edge-Detection and Filtering Framework

All HSL point clouds at different wavelengths share identical 3-D spatial coordinates, recording the hyperspectral reflectance characteristics of the scanned position. In this study, we propose a hyperspectral point cloud edge-effect detection and filtering framework that integrates histogram of intensity distribution, edge detection, and spherical-spatial filtering (HIDEDaSF) processes. The algorithm framework consists of four main steps: 1) histogram construction of intensity distribution; 2) rough edge detection; 3) refined edge detection; and 4) spherical-spatial filtering. The

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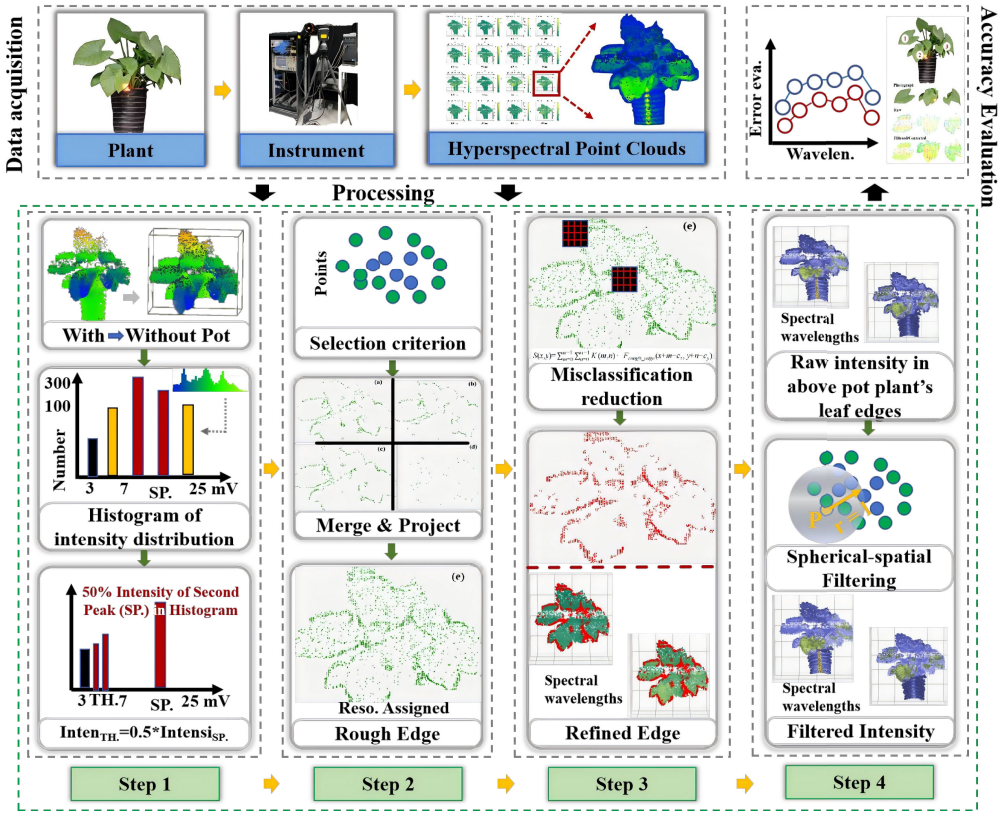


Fig. 4. Overall workflow chart for HIDEDaSF processing framework.

overall workflow of the proposed framework is illustrated in Fig. 4.

1) Mapping Histograms of Intensity Distribution Across Different Wavelengths: To clearly and specifically illustrate the differences in point cloud intensity characteristics across different wavelengths obtained from the HSL, intensity distribution histograms were generated for all 32 spectral wavelengths (Fig. 5). The pot background was removed, and only leaf regions associated with biochemical parameters were retained. We uniformly selected 16 representative wavelengths corresponding to those leaf regions. It seems to be an optional reference to determine an optimal threshold for edge detection based on the abundant spectral intensity distribution patterns for the HSL system. Thereby, in preliminary tests, several strategies for generating edge thresholds were evaluated, including using the mean intensity value of the bin with the highest frequency in the histogram as the threshold, and using the mean of the bin with the highest frequency among those with intensities greater than zero. However, because of background noise, these approaches did not consistently yield stable or reliable separation results across all spectral wavelengths. Finally, we selected half of the mean intensity value of the second-highest frequency nonzero intensity bin as the threshold for each wavelength, denoted as $I_{threshold,\lambda}$. with red line and TH. Value (shown in Fig. 5). This strategy effectively avoids the first peak, which may typically correspond to background noise or large low-reflectance regions, and provides more stable detection of edge point clouds.

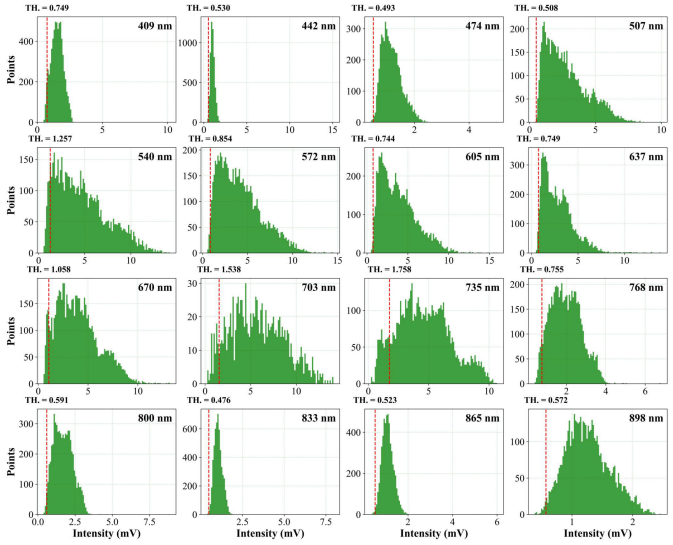


Fig. 5. Histograms of intensity distribution across different wavelengths for the HSL system. TH represents the threshold value of the edge in the following rough detection.

It also avoids classifying too many point clouds with complete light spot energy as edge point clouds.

2) Rough Edge-Detection Result Across Different Wavelengths: Based on the intensity thresholds determined for each wavelength, hyperspectral point clouds with intensity values greater than zero but lower than the corresponding edge

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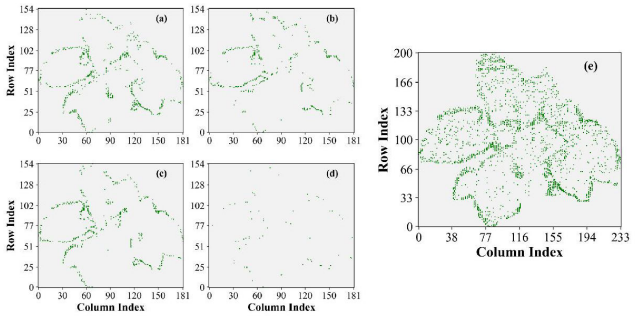


Fig. 6. Rough edge of example wavelengths of (a) 409 nm, (b) 540 nm, (c) 670 nm, (d) 800 nm, as well as final (e) 32 merged rough edge-detection results.

threshold were automatically extracted to form the rough edge-detection results. The selection criterion can be expressed as follows:

$$P_{rough_edge,\lambda}(i) = P_{\lambda}(i), \quad 0 < I_{P_{\lambda}(i)} < I_{threshold,\lambda} \tag{1}$$

where $P_{rough_edge,\lambda}(i)$ is the rough edge points at λ wavelength, $P_{\lambda}(i)$ denotes each point at λ wavelength, and $I_{P_{\lambda}(i)}$ represents the intensity value for the current i_{th} point at λ wavelength.

Through the rough edge-detection process, the approximate boundary points of different wavelengths were obtained, as shown in Fig. 6(a)–(d) (green points) for some example wavelengths. After that, the point clouds from all 32 wavelengths were merged to generate the final rough boundary point cloud, denoted as $P_{rough_edge}(i)$. By projecting the point cloud along the z-axis, the final rough boundary distribution was obtained, as shown in Fig. 6(e) (green points). The rough image $F_{rough_edge}(x, y)$, 0/1 binary value image can be expressed as

$$\mathrm{F_{rough_edge}(x, y)} = \begin{cases} 1, & \text{if edge point in postion (x,y),} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

As shown in Fig. 6(a)–(e), the rough detection results may not always capture the edge point for each wavelength; however, the combined result across all 32 spectral wavelengths, as illustrated in Fig. 6(e), effectively and absolutely encompasses nearly all potential edge-effect points, which also illustrates the spectral advantage regarding HSL data processing.

3) Refined Edge Detection: As shown in Fig. 6(e), during the rough detection process, some points with relatively low echo intensity inside the leaf were incorrectly identified as edge points. This misclassification may occur because the lower intensity values in these regions are associated with reduced biochemical constituent concentrations of the vegetation. To remove these falsely detected points, a refined edge-detection procedure was implemented.

In this step, the original rough edge points detection results in Section II-C2 were projected into a grid space with a spatial resolution approximately equal to the LiDAR sampling resolution. In particular, the point cloud was vertically projected along the z-axis to generate a 2-D image with a spatial resolution of 0.45 cm, where the pixels containing laser points were specially assigned a value of 1 and others were assigned 0 (like in Fig. 6). Simultaneously, an auxiliary array was created to record, for each grid cell, the spatial coordinates and the 32 wavelength intensity indices of all contained points. A 4×4 convolutional window was then applied to the binary image. When more than four (including four) pixels within the window had values of 1 and the current pixel also had a value of 1, the current pixel was classified as a valid edge pixel and retained. Otherwise, it was discarded.

In fact, the window length value of 4 we set was not chosen randomly. It was determined through experiments that if the setting is too small, many points at the edge will be missed because it cannot meet the condition that the gray values of the 2×2 pixel group are all 1, that is, it is impossible for there to be four edge pixels simultaneously present. Since our condition is to make the sum of pixel values within the sliding window space as small as 4, 2×2 set is too strict. When the setting is $6^{*}6$, there are too many pixels (at least it is easy to be greater than 4) with a gray value of 1 in a 6×6 space, which will introduce a lot of noise points, and thus it increases a lot of false edge points. After repeated verification, we found that when the resolution is 0.45 cm, a 4×4 sliding window is the most suitable, and it retains the most original and accurate number of edge points.

By traversing the entire image, all valid edge pixels were extracted to produce the final fine edge point results. The following detailedly describes the process:

By constructing a w $\times$ w window-sized convolution kernel where w= 4 herein, each kernel element value is 1, the convolution kernel can be assigned as

$$\mathrm{K(m, n)} = 1, \quad \mathrm{m, n} = 0, 1, \ldots, \mathrm{w} - 1. \tag{3}$$

The neighborhood convolution sum $S(x, y)$ for each pixel F_{rough_edge} was calculated as

$$\mathrm{S(x, y)} = \sum_{\mathrm{m}=0}^{\mathrm{w}-1} \sum_{\mathrm{n}=0}^{\mathrm{w}-1} \mathrm{K(m, n)} \cdot \mathrm{F_{rough_edge}}\left(\mathrm{x} + \mathrm{m} - \mathrm{c_x}, \mathrm{y} + \mathrm{n} - \mathrm{c_y}\right). \tag{4}$$

Although we used the 4×4 convolution kernel, users can also use other sized kernel, even if it is not even numbers. The offset of the convolution center is calculated as follows:

$$\mathrm{c_x} = \mathrm{c_y} = \left[\frac{\mathrm{w} - 1}{2}\right]. \tag{5}$$

Here, c_x and c_y represent the horizontal and vertical offsets of the convolution center, respectively. When the kernel size w= 4, both c_x and c_y are equal to 1, indicating that the convolution center corresponds to the second row and second column of the kernel. When the convolution window extends beyond the image boundary, constant padding with a value of zero is applied.

The convolution-derived neighborhood sum $S(x, y)$ represents the 1-value pixel density within the w $\times$ w neighborhood surrounding the current pixel. If the $S(x, y)$ exceeds a predefined threshold T_e (set to 4 in this study) and the current pixel

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is valid and assigned to 1, the current pixel is then classified as a true edge pixel, denoted as $M_{refined_edge}(x, y)$

$$\mathrm{M}_{refined_edge}(\mathrm{x},\mathrm{y}) = \begin{cases} 1, & \text{if } \mathrm{S}(\mathrm{x},\mathrm{y}) \geq \mathrm{T_e} \text{ and } \mathrm{F}_{rough_edge}(\mathrm{x},\mathrm{y}) = 1 \\ 0, & \text{Otherwise.} \end{cases} \quad (6)$$

Considering that the edge region may appear relatively thick due to the laser points scanning, the edge image $M_{refined_edge}$ can be further dilated once more if necessary. Based on the previously recorded point cloud index, the image was back-projected and transformed into the refined edge point cloud set, denoted as $\mathrm{P}_{refined_edge,\lambda}(\mathrm{i})$

$$\mathrm{P}_{\mathrm{refined_edge},\lambda}(\mathrm{i}) = \left\{ \left(\mathrm{X_i}, \mathrm{Y_i}, \mathrm{Z_i}, \mathrm{I}_{\mathrm{P}_\lambda(\mathrm{i})}\right) \middle| \left(\mathrm{x_i}, \mathrm{y_i}\right) \in \mathrm{M}_{\mathrm{refined_edge},\lambda}(\mathrm{x},\mathrm{y}) = 1 \right\}. \quad (7)$$

Method 1 Refined_Edge_Detection in HIDEDaSF

Input : rough_edge_points.csv, resolution, windowLength, edgeThresholdNm
Output : refined_edge_points.csv
Begin
Load point cloud data from all_edge_points.csv → DF(DataFrame)
Extract columns [X, Y, Z, distance, wavelength_409, . . . , wavelength_914]
Define rough projection grid
X_range ← [min(X), max(X)]
Y_range ← [min(Y), max(Y)]
x_bins ← ceil((X_range.max - X_range.min) / resolution)
y_bins ← ceil((Y_range.max - Y_range.min) / resolution)
Initialize binary projection[y_bins, x_bins] ← 0
For each point P in DF:
x_idx ← floor((P.X - X_min) / resolution)
y_idx ← floor((P.Y - Y_min) / resolution)
projection[y_idx, x_idx] ← 1
Record mapping (y_idx, x_idx) → point indices
Save rough projection image (rough_edge_projection.png)
Perform neighborhood detection
kernel ← ones (windowLength, windowLength)
neighbor_sum ← convolve(projection, kernel)
refined_mask ← (neighbor_sum ≥ edgeThresholdNumber) ∧ (projection == 1)
Save (and Dilate) refined projection image (refined_edge_projection.png)
Extract refined edge points:
refined_indices ← indices of points whose cells satisfy refined_mask == 1
refined_DF ← DF[refined_indices]
Save refined_DF → refined_edge_points.csv
End

To facilitate understanding of the above procedure, the pseudocode of this step is given as follows.

4) Spherical-Spatial Filtering for Edge Points Regarding HSL System: Through step Section II-C3, the point cloud edges were accurately detected, resulting in a refined edge point cloud dataset (see Section III). These final refined edge points are shared across all spectral wavelengths. In this step, spherical-spatial filtering is applied to these edge points to eliminate the associated edge effects. In particular, for each edge point, all nonedge points within a spherical neighborhood of radius r are identified first, and then the mean intensity of these neighboring leaf points is assigned as the final filtered intensity value of the current edge point. Here, we suggest that users take 10–20 times the distance between an edge point and its nearest adjacent edge point as the radius. The distance radius varies under different point cloud densities, and it should be determined according to the specific situation. This procedure is iteratively performed for all spectral wavelengths to complete the spherical-spatial filtering process.

D. Evaluation Indicators

To evaluate the performance of the proposed edge-effect detection and filtering framework before and after filtering, three statistical parameters were employed, including standard deviation σ, CV, and the ratio of CV ε [17]. The σ represents the fluctuation and divergence of the intensity data. But it is not appropriate to verify the difference if the data have different numerical scales. The CV is then taken to eliminate the scale influence, and it is expressed as

$$\mathrm{CV} = \sigma/\mu \quad (8)$$

where μ is the average value of intensity data. Besides, the ratio of the CV is employed to assess the effectiveness of the detection and filtering, which is defined as

$$\varepsilon = \mathrm{CV_{cor}}/\mathrm{CV_{raw}} \quad (9)$$

where CV_{cor} is the coefficient of the variation of the edge-effect-corrected intensity and CV_{raw} is the coefficient of variation of the raw recorded intensity of points. If the parameter of ε is less than 1, it means the filtering model gains a nice performance and filtering is effective. The less the ε value is, the better the filtering of the framework performs

III. Results

A. Final Refined Edge Detected Results

After the refined edge detection, a more complete and accurate leaf-edge region was obtained, as shown in Fig. 7. At this stage, it is evident that, compared with the rough edge results in Fig. 6(e), the interior noise points and low-reflectance points within the leaves have been effectively removed, and most pseudoedge points have also been eliminated. For the uppermost leaf, the detected edge appears somewhat thicker than that of the other leaves, but it is reasonable because the incident angle on this leaf is very large. Thus, in the leaf-tip region, the originally circular laser footprint becomes elongated due to oblique incidence, causing the returned energy to scatter into a very big elliptical pattern. As a result, the measured intensity in this region frequently falls below its nominal value, naturally forming an extended edge zone.

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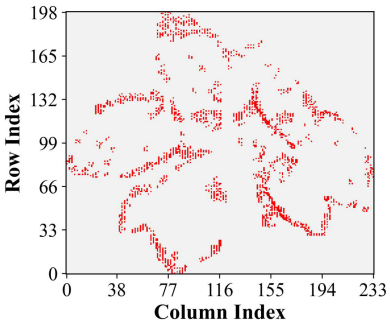


Fig. 7. Refined edge-detection results based on the rough edge in Fig. 6(e), utilizing step Section II-C3.

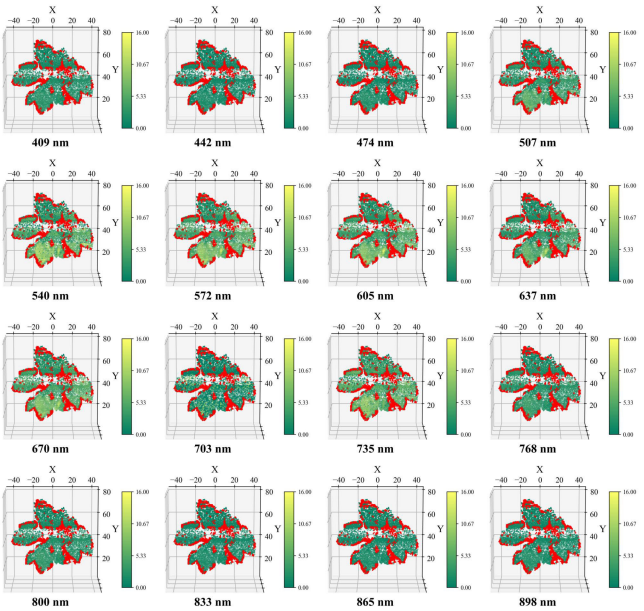


Fig. 8. Leaf intensities at all wavelengths with the refined edge cover.

By overlaying the final refined edge points onto the leaf surfaces across all spectral wavelengths, the intensity-rendered figures for each wavelength were obtained (Fig. 8). It can be observed that the extracted edges align almost perfectly with the actual leaf surfaces. Although some leaf regions exhibit slight jagged discontinuities, these imperfections primarily arise from the limitations in the scanning resolution of the current HSL system, which is not yet capable of achieving highly and ultrafine sampling conditions.

B. Corrected Edge Intensities Across Different Wavelengths With HIDEDaSF

To further visualize and assess the effectiveness of the proposed filtering and correction procedure, the corrected edge intensities obtained after step Section II-C4 were rendered, as shown in Fig. 9. The locations and intensities of the interior leaf points remained unchanged and were not subjected to any modification. But it is evident that, after processing with the HIDEDaSF framework, the edge-effect phenomena are no longer apparent, and the contrast between the edge regions

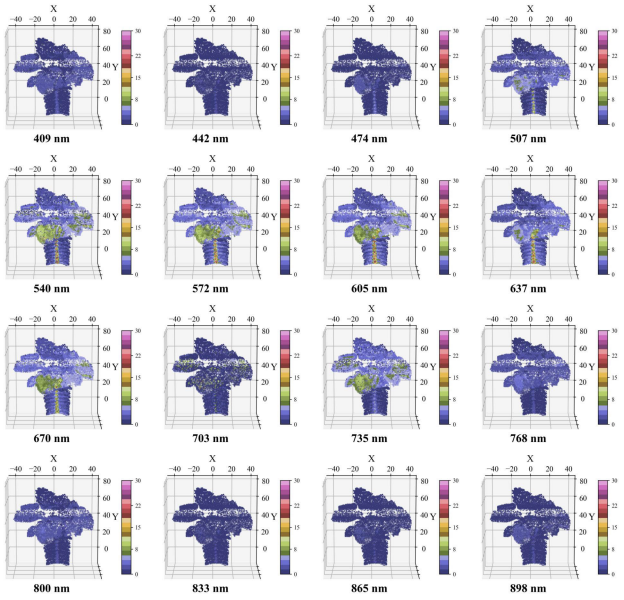


Fig. 9. Edge-filtered or edge-corrected 3-D intensities at all wavelengths with the pot. The reason for choosing the colormap "tab20b" is to differentiate and visualize in detail the intensity and the edge effect for leaves.

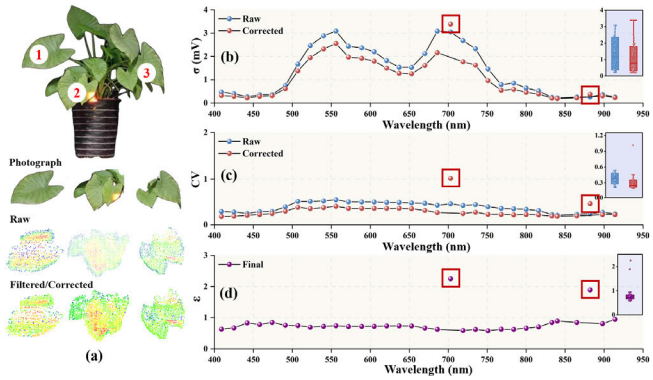


Fig. 10. Visible photographs and quantitative evaluation results. (a) Intensity change for three typical leaf-edge regions, and (b)–(d) standard deviation σ, CV, and ε across different wavelengths, respectively.

and the leaf interior has been substantially reduced. Compared with Fig. 3, the improvement is particularly remarkable in the 540–637-nm spectral range, where the originally pronounced edge-effect artifacts have been significantly weakened or have disappeared altogether. These visualization results clearly demonstrate the effectiveness of the proposed algorithm in detecting and correcting edge effects in HSL vegetation point clouds.

C. Quantitative Analysis of the Performance for Edge-Effect Filtering

To quantitatively evaluate the performance of the edge-effect filtering framework, three statistical indicators, including standard deviation σ, CV, and the ratio of CV ε, were employed to assess the intensity variability. For the raw point cloud, three representative leaf-edge regions exhibiting pronounced edge effects were selected, as shown in Fig. 10(a). As shown

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TABLE I
AVERAGE OF THREE ERROR INDICATORS ACROSS ALL THE WAVELENGTHS FOR THE RAW AND CORRECTED INTENSITY DATA

Indicators	σ (mV)		*CV*		ε
Types	raw	corrected	raw	corrected	final
Values	1.3497	1.0436	0.3774	0.2706	0.7288

in Fig. 10(a), the visualization of leaf-edge effects varies substantially, before and after filtering. The edge points, which initially exhibited noticeably lower intensities compared with the leaf interior, gradually approach the intensity levels of interior points after applying the proposed framework.

The quantitative spectral intensity variation evaluations before and after filtering in these regions were then calculated for all wavelengths [Fig. 10(b)–(d)]. With respect to the σ, it is evident that the corrected intensities show consistently reduced values across all wavelengths compared with the raw ones. It indicates that the proposed method effectively suppresses the excessive intensity fluctuations between edge and interior leaf points. The *CV* results, in Fig. 10(c), further support this conclusion, demonstrating satisfactory filtering and correction performance for the majority of wavelengths. Moreover, the ε for all wavelengths is less than 1, fundamentally validating the effectiveness of the proposed framework and confirming its capability to correct edge-effect distortions in HSL's vegetation point clouds.

It is worth noting that, at the beginning and end of the spectral range, the corrected standard deviation and *CV* are not consistently lower than the raw values. The fact is that these wavelengths and 703 nm (highlighted by the red box) correspond to low SNR regions of the current HSL system, as clearly reported in our previous study [1], [27]. It can be seen that these related abnormal values fall outside the whisker range in the boxplots and do not influence the overall conclusion of this study. For all other wavelengths with higher SNR and better data quality, the proposed framework still and always achieves excellent filtering and recovery performance.

Table I summarizes the mean values of the three statistical error indicators for the raw and corrected intensity data across all wavelengths. As shown in Table I, the corrected standard deviation is 1.0436 mV, representing a 22.68% reduction compared with the raw value of 1.3497 mV. The *CV* decreases from 0.3774 to 0.2706 after filtering, corresponding to a 28.30% reduction. The substantial decreases in both standard deviation and *CV* confirm the reliability of the proposed edge-effect filtering framework. In addition, the ε of 0.7288 further demonstrates the stability and effectiveness of the algorithm framework across the spectral range.

IV. DISCUSSION

This study presents, for the first time, a complete framework for detecting and correcting edge effects in hyperspectral point cloud data, with a fundamental focus on detecting and an intensity recovering algorithm. As one of the three major radiometric effects in LiDAR measurements, the edge effect severely restricts the consistency of echo intensity from fine vegetation substructures such as leaves. Previous ad hoc treatments, either ignoring or discarding edge-affected points [15], [25], [26], do not offer a viable solution for HSL processing. Instead, they further increase the difficulty of quantitative analysis and applications relying on accurate intensity, such as the retrieval of chlorophyll and other biochemical traits, thereby limiting the development of quantitative and fine-scale HSL remote sensing. To support the advancement of HSL quantitative remote sensing [20], research on the generation of physical mechanisms, detection methods, and filtering strategies for edge effects is urgently needed. This HIDEDaSF framework proposed herein represents an important supplement to existing radiometric correction theories and techniques. Compared with previously reported studies [22], this work constitutes the first systematic, comprehensive investigation, analysis, detection, and filtering of edge effects on HSL data. Overall, this study significantly broadens the knowledge base and current scope of HSL research.

Regarding the edge effect, it arises from the loss of echo energy when the laser footprint partially intersects the leaf boundary. In this study, the phenomenon is clearly demonstrated in vegetation applications. Figs. 3 and 8 reveal an interesting but physically reasonable observation that edge-effect points do not necessarily correspond to the single outermost layer of points. During the period, during which the laser footprint gradually moves from the leaf surface toward the exterior and eventually lands entirely on background objects, multiple scanning points are generated along this transition. Consequently, the edge region contains several spatial strata of points rather than a single layer. Each stratum represents a ring of edge points, in which only a certain fraction of the laser energy is returned from the leaf surface, while the remaining portion interacts with the surrounding background. The number of edge-effect points and the apparent thickness of the edge region depend on multiple factors such as beam divergence angle, measurement distance, leaf spatial geometry, and scanning angular resolution of the HSL system.

Based on the refined edge-detection results illustrated in Fig. 8 and the postfiltering intensity visualization in Fig. 9, the proposed edge-effect detection and filtering algorithm framework demonstrates reliable performance. The HIDEDaSF framework accounts for the spectral variation in point cloud distribution and intensity patterns across wavelengths and achieves robust edge detection that remains valid for all wavelengths. From a quantitative perspective, the results in Fig. 10 and Table I clearly indicate that the filtering reduces both the intensity standard deviation and the *CV* values within leaf regions beyond 20%. Furthermore, the ratio of the CV ε always remain less than 1 herein for all the wavelengths with a final average value of 0.7288, which further confirms the validity and stability of the proposed framework.

Although this study represents the first effort to detect and correct edge effects in hyperspectral point cloud data, thereby expanding the rapidly evolving research direction of HSL, the current correction strategy is still fundamentally

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based on filtering. It should be acknowledged that the proposed method has limitations and room for improvement. In particular, physically accurate restored intensities ultimately require a mechanistic correction approach rather than an empirical solution. In future work, we will focus on developing a physically-based model for edge-effect correction of hyperspectral point clouds and further investigate the compensation strategies for different target types. These advances are expected to enhance the quantitative capability of HSL and promote its application ability in accurate and fine-scale 3-D vegetation remote sensing. Furthermore, it is necessary to humbly and sincerely point out that, given the current capabilities of the HSL, apart from the testing targets in the laboratory, in the future, more field experiments with much higher acquisition speed equipment will be needed to evaluate the stability and reliability of the HIDEDaSF algorithm.

V. Conclusion

This study presents the first comprehensive investigation, analysis, detection, and filtering of edge effects on HSL point cloud data, since the emergence of HSL technology. We examined the physical mechanisms responsible for edge-effect generation and developed a new integrated algorithmic framework, termed HIDEDaSF, which combines histogram-based intensity distribution analysis, edge detection, and spherical-spatial filtering. Experimental results demonstrate that the proposed framework can robustly identify edge-affected points across all wavelengths of the hyperspectral point clouds. After filtering, the standard deviation and CV of leaf-edge intensities are reduced by 22.68% and 28.30%, respectively, indicating a substantial decrease in intensity discrepancies between edge and interior leaf regions. In addition, the mean ratio of CV across all wavelengths is 0.7288, less than 1, confirming the validity and stability of HIDEDaSF for mitigating edge effects throughout the spectral range. Although tailored to our HSL system, the proposed framework can also be practical and applicable to other multispectral and hyperspectral laser scanning platforms. The findings of this work enrich the fundamental knowledge of HSL radiometric behaviors, significantly expand the research scope, and facilitate methodological development of HSL remote sensing.

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